

The AI Trust Graph: A HITS-Based Authority Model for Trust Propagation in AI-Mediated Recommendation and Commerce

Working Paper

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Abstract

Large language models increasingly act as the first point of discovery for products, vendors, and services — a role search engines held for two decades. Yet the mechanism by which an LLM comes to “trust” one entity over another is opaque, and existing Generative Engine Optimization (GEO) tools measure *outcomes* (how often a brand is mentioned) without modeling the *structure* that produces those outcomes. This paper defines the **AI Trust Graph** theory — a bipartite source→entity corroboration graph scored with a HITS-style (Hyperlink-Induced Topic Search) hub-authority algorithm — and then scopes a concrete, buildable-in-hours artifact that tests the theory’s core empirical claim directly against live LLM behavior, with no crawling, no graph database, and no external data dependency: a **Recommendation Stability Tool** that queries multiple production LLMs with paraphrased prompts and reports a Trust Score, a stability index (RSI), and a head-to-head competitor ranking for any named entity and category. Section 6 is the buildathon deliverable; everything before it is the theory that explains *why* the number the tool produces means something.

1 Motivation: The New Discovery Layer

For twenty-five years, discoverability was mediated by a single, well-understood structural fact: the web is a directed graph of hyperlinks, and PageRank [1] converts that graph into an authority ranking. An entire discipline — SEO — grew around influencing one’s position in that graph.

That structural fact no longer describes how a large share of discovery happens. A user asking an LLM “what’s the best payment gateway for a D2C startup in India” receives a synthesized answer drawn from an opaque mixture of pretraining data, retrieval-augmented context, and RLHF-shaped preferences. There is no hyperlink to click, no crawlable graph to rank against, and — critically — no consensus yet on *what the analogous structure even is*.

A cottage industry (Profound, Otterly.ai, Peec AI, and similar GEO tools) has emerged to measure *whether* a brand gets mentioned by LLMs. This is useful, but it answers a narrower question than the one that matters. It tells a brand where it stands; it does not explain *why* the ecosystem trusts what it trusts, and it offers no transferable theory of how trust propagates from sources to entities across the whole market.

The central claim of this paper. AI models do not recommend entities in proportion to how often they are mentioned on the web. They recommend in proportion to the *authority-weighted, recursively propagated* corroboration behind an entity — the same structural insight PageRank applied to hyperlinks, applied instead to a source→entity corroboration graph. Mention count is a vanity metric; weighted, propagated authority is the real signal.

This paper is deliberately framed at two levels simultaneously, because conflating them is the most common failure mode of ideas in this space: (1) a **measurement system** — can we quantify trust propagation empirically and validate it against real model behavior? — and (2) an **economic infrastructure thesis** — if the measurement is valid, who owns the resulting standard, and why does it compound rather than get commoditized? We treat (1) as the buildathon deliverable and (2) as the reason the idea is worth building in the first place.

2 Why HITS, Not PageRank

Pure PageRank [1] assumes a **unipartite** graph: pages link to pages, and authority is derived entirely endogenously, with no outside input required. The web-of-links case is special precisely because every node is the same type of thing.

The source→entity corroboration graph is different in kind: sources (Wikipedia, trade press, review sites, forums, blogs) point at entities (brands, products, vendors), but sources rarely cite each other in an easily crawlable way, and entities do not cite anything at all. This is a **bipartite** structure, and the correct algorithmic ancestor is not PageRank but Kleinberg’s HITS algorithm [2] — Hyperlink-Induced Topic Search — originally designed for exactly this hub/authority split.

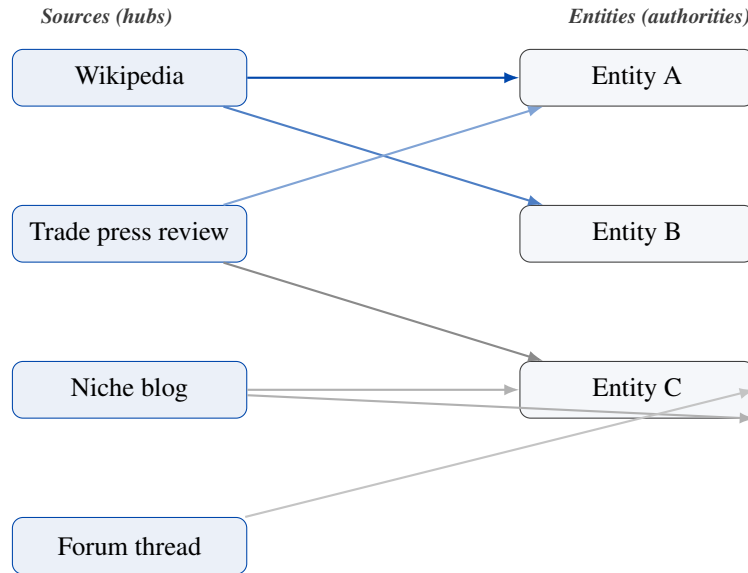


Figure 1: The bipartite corroboration graph. Wikipedia (high hub authority) casts a small number of high-weight votes; the forum (low hub authority) casts many low-weight votes at Entity C. Entity A accumulates the highest authority despite the fewest total mentions — mirroring why link farms never worked against real PageRank.

In HITS, a source’s hub value depends on the quality of the authorities it points to, and an entity’s authority value depends on the quality of the hubs pointing to it — a mutually recursive definition resolved by iterating to a fixed point, in the same spirit as PageRank’s eigenvector solution.

3 Formal Model

3.1 The Corroboration Graph

Let S be the set of source nodes and E the set of entity nodes. The corroboration graph is

$$G = (S \cup E, W), \quad w(s \rightarrow e) \in \mathbb{R}_{\geq 0},$$

where an edge $s \rightarrow e$ exists whenever source s mentions, cites, reviews, or otherwise corroborates entity e , and $w(s \rightarrow e)$ encodes both the source’s authority and the strength of corroboration (a passing mention vs. a detailed review vs. a specific factual citation).

3.2 HITS Iteration

Let $h(s)$ denote hub score and $a(e)$ denote authority score. The mutually recursive update is

$$a(e) \leftarrow \sum_{s:s \rightarrow e} w(s \rightarrow e) \cdot h(s), \quad (1)$$

$$h(s) \leftarrow \sum_{e:s \rightarrow e} w(s \rightarrow e) \cdot a(e), \quad (2)$$

with $\mathbf{a}$ and $\mathbf{h}$ renormalized to unit norm after each iteration. This converges to the principal eigenvectors of $WW^\top$ and $W^\top W$ respectively [2].

3.3 Recommendation as a Conditional Distribution

A deeper and non-obvious point must be made precise before any of the above is useful: an LLM does not have a stored “trust value” for an entity. It has a conditional probability distribution over entities given a query and a model. Formally,

$$R : Q \times M \rightarrow \Delta(E),$$

where Q is the query space, M is the model space, E is the entity space, and $\Delta(E)$ is the probability simplex over entities. For a query class c with query distribution D_q , we define the empirically measurable

$$\text{Trust}(e, c, m) = \mathbb{E}_{q \sim D_q} [P(e \text{ recommended} \mid q, m)].$$

This has an immediate and important consequence: **trust is not a scalar attached to an entity — it is a high-dimensional object indexed by query class and model.** A payment gateway can have near-total trust centrality for “UPI integration for Indian SaaS” and near-zero trust for “payments for EU marketplaces.” Any single collapsed score is, by construction, throwing away the information that makes the measurement useful. The graph in Section 3 should therefore be understood as one slice, G_c , of a family of graphs indexed by query class c .

3.4 Recommendation Stability Index (RSI)

Because the distribution above is measured empirically over paraphrased queries, we define stability as inverse variance across semantically equivalent phrasings:

$$\text{RSI}(e, c, m) = 1 - \frac{\text{Var}_{q \sim D_q} [P(e \text{ recommended} \mid q, m)]}{\text{Max_Variance}}.$$

High RSI means the model recommends the entity consistently regardless of how the query is phrased; low RSI flags a fragile, prompt-sensitive recommendation — a useful early-warning signal distinct from raw recommendation frequency.

3.5 AI Trust Centrality (AITC)

Cross-model consensus is aggregated as

$$\text{AITC}(e, c) = \sum_{m \in M} w_m \cdot \text{Trust}(e, c, m),$$

where w_m is a model weight (e.g. market share). A stronger formulation uses pairwise Pearson correlation of per-model trust scores as an edge weight in a model $\times$ entity bipartite graph, so that entities trusted *consistently across* GPT, Claude, and Gemini score higher than entities that are a favorite of exactly one model family.

3.6 The Seeding Problem — Stated Plainly

Pure PageRank needs no external input: pages link to pages, all the way down. The corroboration graph cannot do this cleanly, because sources do not cite each other in an easily observable way. **Source authority $h(s)$ must therefore be seeded from outside the graph.** We rank the available seeding strategies by rigor, not convenience:

- (1) **Manual tiering** (fastest to ship): Tier 1 — Wikipedia, major outlets; Tier 2 — established trade press; Tier 3 — blogs, forums, UGC.
- (2) **Third-party domain-authority proxies**: Moz DA, Ahrefs DR, or similar, used as an initial hub-score prior.
- (3) **A secondary source-to-source citation graph**: does the trade report cite Wikipedia? Does the blog cite the trade report? Run *actual* PageRank on this smaller, genuinely unipartite graph to generate authority priors, then propagate those priors into the bipartite entity layer.

Being upfront about this dependency — rather than presenting the model as a clean, self-contained PageRank analogue — is what makes the model credible rather than hand-wavy. It is also, not coincidentally, the same limitation every citation-based rating agency (Moody’s, S&P) resolves by combining public methodology with proprietary, partially manual, initial calibration.

4 Validation: How You’d Know It’s Real

A graph-theoretic score is only interesting if it predicts something outside itself. We propose a direct, falsifiable protocol:

1. Compute $\text{AITC}(e, c)$ for a basket of entities from the source-entity graph, independent of any LLM.
2. Separately, query production LLMs (GPT-4o/5, Claude, Gemini) with a template of paraphrased prompts per category and empirically measure how often and how confidently each entity is recommended — this yields $\text{Trust}(e, c, m)$ directly, with no reference to the graph.
3. Correlate the two. A statistically significant positive correlation across categories and models is the empirical result the entire thesis rests on. A null result means the model needs rethinking *before* anything commercial is built on top of it.

This step is what separates the AI Trust Graph from a plausible-sounding metaphor. Everything upstream (Sections 3–4) is a hypothesis about structure; this step is the experiment that tests it, and it

is buildable with API access alone — no model weights, no proprietary retrieval logs, no institutional access required.

5 What Is Measurable and What Is Not

Measurable today	Not directly measurable
Recommendation frequency across paraphrased query variants	Causal mechanism (<i>why</i> the model recommends something)
Cross-model agreement on top- k recommendations	Training-data attribution without model access
Temporal stability across model versions (longitudinal)	Pretrain / fine-tune / RLHF contribution split
Source correlation via RAG citation logs, where exposed	Ground-truth internal “trust” representation (none exists)

Table 1: The honest measurement boundary. We are measuring the *behavioral shadow* of a latent structure, not the structure itself — the standard epistemic condition of behavioral science, not a flaw unique to this approach.

6 What Gets Built, Right Now

No graph database. No web crawling. No source-authority seeding. Section 4’s theory justifies *why* the number matters; none of its machinery needs to run live for the buildathon. The buildable artifact is a single, self-contained pipeline that goes from a text input to a live dashboard in one request cycle.

6.1 The Product, in One Sentence

Type in your brand, your category, and 2–3 named competitors. The tool fires paraphrased queries at 3 live LLMs, checks who each model actually recommends, and shows you a Trust Score, a stability rating, and a head-to-head win rate against each competitor — per model and combined.

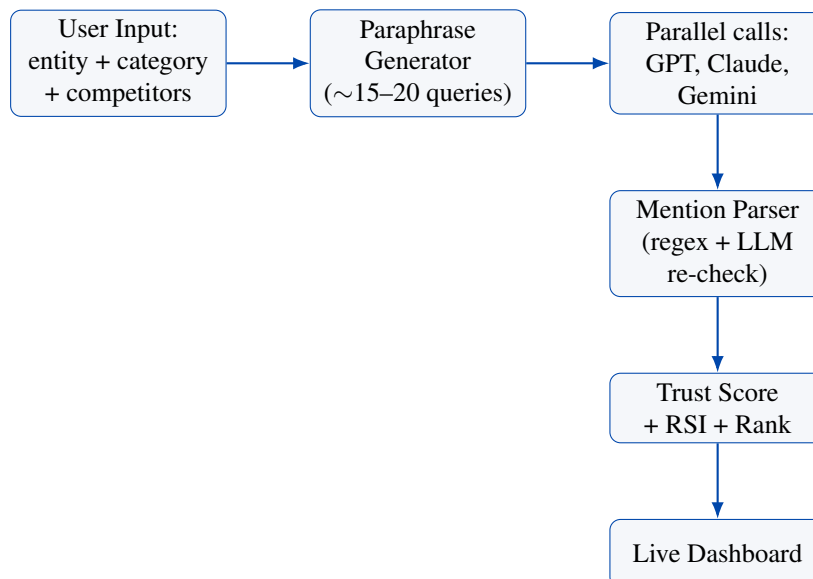


Figure 2: The entire pipeline runs synchronously per demo query — no background jobs, no persistent crawl, no stored graph. Everything left of the dashboard is stateless.

6.2 Exact Scope for the Build Window

- **Input form:** entity name, category/use-case string, 2–3 competitor names. That’s the entire input surface.
- **Paraphrase generation:** one LLM call that takes the category string and returns 15–20 differently-phrased natural queries (“best X for Y”, “I need X, what should I use”, “compare top X tools”, with varied formality and constraints). Cache this per category so re-runs are cheap.
- **Multi-model firing:** send every paraphrase to 3 provider APIs in parallel (OpenAI, Anthropic, Google — whichever three you have keys for; two is a fine fallback if a key isn’t available at demo time). This is ~50–60 total API calls per full run — a few seconds with concurrency, not minutes.
- **Mention parsing:** for each response, check (a) does it name the target entity, (b) does it name each competitor, (c) rough position/emphasis (mentioned first vs. buried, recommended vs. merely listed). A simple regex/fuzzy-match pass plus one structured-output LLM call (“return JSON: which of these names appear and in what order”) is enough — this is the one place worth spending real build time, since it is the direct implementation of Section 4.3’s extraction problem.
- **Scoring:** $\text{Trust}(e, c, m) = \frac{\# \text{ responses recommending } e}{\# \text{ total paraphrases}}$ per model, exactly as defined in Eq. (3); RSI per Eq. (5) computed as $1 - \text{normalized variance of that indicator across paraphrases}$; a simple head-to-head win rate against each named competitor.
- **Dashboard:** one screen — a bar per model showing Trust Score for you vs. each competitor, an RSI badge (“stable” / “fragile”), and one sentence: “*You’re recommended in X% of responses across 3 models, beating {competitor} Y% of the time.*”

6.3 What This Deliberately Skips

No source-corroboration graph, no HITS iteration, no web-authority seeding, no longitudinal storage, no source attribution. Those stay in the paper as the theory that explains why a cross-model, paraphrase-averaged score is the right thing to measure at all — they are not on the buildathon critical path, and pretending otherwise the day of the demo is how nothing ships.

6.4 Relevance to Razorpay

The demo category should be chosen to land with the judges: run it live on “best payment gateway for an Indian D2C startup” with Razorpay as the target entity against 2–3 named competitors. The pitch in one line: *as AI agents start choosing which vendor to pay, this is the instrument that tells a payments platform whether it’s actually the one getting chosen — and how fragile that position is.*

7 Frontend Differentiation: Making the Fragility Surface Visible

A backend that computes Trust Score, RSI, and head-to-head rank is necessary but not sufficient. If the frontend renders that output as a bar chart per model, the product is visually indistinguishable from Profound, Otterly.ai, or Peec AI — an aggregate percentage is an aggregate percentage regardless of the theory behind it. The differentiation argued for in Sections 3–5 has to be load-bearing in the UI, not just in the scoring

code. This section specifies the interface precisely enough to hand to a builder — human or agentic — as a build spec, not a mood board.

7.1 The Core UI Decision

The hero of the page is a grid, not a number. Rows are the paraphrased queries (Section 6’s ~15–20 phrasings); columns are the models; each cell is colored by outcome. This is the direct visual implementation of RSI: instead of reporting “62% recommended” as a single figure, the grid lets a viewer *see* the exact subset of phrasings where the entity wins or loses — for example, a visible red column-band across every “cheapest” or “lowest fee” phrasing while every “startup” or category-specific phrasing stays green. That pattern is the finding; a bar chart cannot show it because it has already been averaged away.

7.2 Four-State Cell Encoding

Each grid cell must encode one of four states, not a binary hit/miss — collapsing to two states throws away the distinction between “you won outright” and “you were merely present alongside a competitor,” which is exactly the vanity-metric collapse this paper argues against in Section 1.

State	Meaning
Win	The target entity is recommended in that model’s response to that phrasing; no named competitor is.
Loss	A named competitor is recommended instead of the target entity.
Both	Target entity and at least one competitor are both mentioned/recommended in the same response.
Neither	Neither the target entity nor any named competitor appears in the response (the model answered off-axis, e.g. recommending a category of solution rather than a named vendor).

Table 2: The four-state cell encoding. Distinguishing Both from Win, and Neither from Loss, is what keeps the grid diagnostic rather than a repainted mention-rate chart.

Each cell is clickable. Clicking opens an evidence panel showing the actual model response text that produced that cell’s classification (highlighting the entity/competitor names it matched on). This is the second load-bearing difference from existing GEO tools: the score is inspectable, not just asserted. A viewer who doubts a red cell can read the sentence that caused it.

7.3 Consensus, Rendered as Composition, Not Averaged Away

Above the grid, render a single composition bar — not per-model cards side by side — split into three segments: the fraction of phrasings where all models unanimously recommend the entity, the fraction that split (models disagree), and the fraction of unanimous losses. This is the direct visualization of AITC’s cross-model-consensus claim from Section 4.4: an entity favored consistently across model families is a structurally different (and more defensible) fact than an entity with a high score on one model and a low score on another that happens to average out to the same number. Three separate per-model bar charts do not communicate this; a single composition bar does.

7.4 Head-to-Head, Not Just Self-Score

For each named competitor, render a normalized win-rate bar per model (target vs. that specific competitor, on phrasings where at least one of them appears). This answers the question a marketing or product lead

actually asks — “do I beat Competitor X” — rather than the vaguer “how often am I mentioned in the abstract” that generic GEO dashboards default to.

7.5 Interface Design Tokens (for a builder to follow exactly)

To avoid the generic SaaS-dashboard look (rounded cards, soft drop shadows, cream-and-terracotta or black-and-neon defaults), the reference implementation uses an instrument-panel visual language appropriate to a diagnostic tool rather than a marketing dashboard:

- **Palette:** deep ink-teal background (#0B1413), panel surface (#10201E), hairline borders (#21403C). Four semantic colors carry all meaning: win = amber (#E7A23E), loss = rose (#C05B72), both = teal (#3FAE9F), neither = muted slate (#24322F). No color is decorative; every color on the page is a data state.
- **Type:** a geometric display face (Space Grotesk) for headings, a plain humanist sans (IBM Plex Sans) for body copy, and a monospace face (IBM Plex Mono) reserved specifically for displayed data — query strings, percentages, scores — since that is real tabular data, not a decorative label.
- **Layout:** left-aligned, dense, table-first. The grid is a real `<table>`, not a card grid, because the content is literally tabular (query × model) and a card layout would obscure the row/column structure that makes the pattern visible.

7.6 Implementation Note

A working single-file HTML/CSS/JS reference of this exact interface (mock data, functioning click-to-evidence drawer, functioning head-to-head bars) accompanies this paper as `verdict.html` and is intended to be handed directly to a coding agent (e.g. Antigravity, Claude Code) as the frontend target: replace the hardcoded `queries` and `competitors` arrays with the JSON emitted by the Section 6 pipeline and the interface requires no further redesign.

8 The Moat, Stated Precisely

- **Longitudinal, cross-model behavioral data collected before anyone else thought to collect it.** How does an entity’s recommendation probability shift across model generations? Which entities hold consensus across GPT, Claude, and Gemini versus being model-specific? This dataset does not exist yet, and becomes harder to reconstruct retroactively the longer collection is delayed.
- **Neutral third-party credibility.** OpenAI cannot credibly rate Google’s recommendation bias, and vice versa. An independent measurement layer occupies a structurally defensible position — the Moody’s/S&P position: credible partly *because* it is not one of the entities being rated.
- **Methodology-open, data-proprietary.** Publishing the scoring methodology (query templates, formulas, calibration procedure) as open standard builds legitimacy without giving away the compounding asset, which is the accumulated longitudinal dataset itself.

This is explicitly *not* the same category as brand-specific GEO/“fix my AI visibility” services. Those optimize one entity’s position and are, structurally, a service anyone can replicate. This proposal models the topology of the whole ecosystem — the PageRank-versus-SEO-consulting distinction.

9 Risks and Honest Limitations

1. **The graph may not be a graph at all.** PageRank worked because an explicit, crawlable hyperlink graph existed. The corroboration graph proposed here is a *latent* structure inferred from model outputs, not an observed one. Every claim about “the graph” is really a claim about an inferred statistical structure with non-trivial uncertainty — scientifically valid, but harder to operationalize as infrastructure than the PageRank analogy suggests at first glance.
2. **Non-stationarity is structural, not incidental.** Every model update potentially reshapes the trust topology; scores from one model generation say little about the next. This is a business-model constraint (continuous re-measurement, not a database built once) as much as a technical one.
3. **Goodhart’s Law is the deepest risk.** The moment this metric is widely adopted by marketing teams, it stops measuring authority and starts measuring “ability to perform well against this measurement protocol” — precisely what happened to PageRank (link spam) and Domain Authority (link purchasing). Partial mitigations — continuous methodology evolution, anomaly detection, multi-layer measurement, ground-truth anchoring against independent human review data — reduce but do not eliminate this risk.
4. **Timing risk.** The infrastructure-scale opportunity assumes AI agents meaningfully influence or execute purchasing decisions at scale. That may be true by 2030–2032 and not by 2026. The buildathon-scoped MVP (Section 7) is deliberately designed to be useful under today’s, purely informational, usage pattern — the infrastructure vision is the reason to keep building, not a precondition for the first version to be valuable.

10 Conclusion

The web had an explicit graph to rank; AI-mediated discovery does not, and pretending otherwise is the fastest way to build something that sounds rigorous but is not. The contribution of this paper is to state the correct structural analogy precisely — bipartite corroboration graph, HITS-style hub/authority propagation, not naive PageRank — to formalize the quantities that make the model testable (RSI, AITC), to name the seeding dependency and the Goodhart’s-Law risk explicitly rather than glossing over them, and to scope an artifact (Section 6) buildable end-to-end within a single hackathon window with nothing more than three LLM API keys. If there’s runway left after the demo works, the natural next step is exactly the validation protocol in Section 5 — run it across more categories and let the correlation, not a pitch, make the case for anything bigger.

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